

RNS Institute of Technology, Bengaluru

Department of Computer Science & Engineering

PROJECT SYNOPSIS

Team Name	Polaris
Problem Title	Digital Platform for Monitoring Startup Incubation Programs
Year	2025
Category	Software
Domain/Theme	General / Others

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Abstract

The rapid growth of startup ecosystems has increased the demand for efficient management of incubation programs that support early-stage ventures. Traditional methods such as spreadsheets, emails, and manual documentation are often inefficient, difficult to scale, and lack transparency when handling large numbers of startups, mentors, and investors. To address these challenges, this project proposes the design and development of a scalable digital Startup Incubation Management Platform that centralizes and automates the management of incubation activities. The platform provides an integrated environment where multiple stakeholders—including startup founders, mentors, incubator administrators, and investors—can collaborate through a single system. It supports key incubation processes such as startup onboarding, mentor assignment, milestone tracking, funding monitoring, event and workshop management, and community networking. The system follows a modular architecture using microservices, enabling independent services for startup management, mentorship management, funding tracking, event coordination, and community interaction. In addition, the platform incorporates analytics dashboards that provide real-time insights into startup performance, program effectiveness, and incubation outcomes. A secure authentication and role-based access control system ensures that different users access only the features relevant to their roles.

By leveraging scalable cloud technologies, modern web frameworks, and structured data management, the proposed solution aims to improve operational efficiency, transparency, and collaboration within startup incubation programs, ultimately supporting the growth and success of emerging startups.

Objectives

1. Develop a centralized digital platform that supports the end-to-end management of startup incubation programs, from onboarding to performance monitoring. 2. Enable efficient collaboration between startups, mentors, incubator administrators, and investors through a single shared system. 3. Provide real-time tracking of startup milestones, growth metrics, and incubation progress through dashboards and reporting tools. 4. Facilitate mentor assignment, session scheduling, and feedback collection to support structured mentorship. 5. Manage incubation cohorts, training sessions, workshops, and demo days through an integrated event management module. 6. Deliver analytics dashboards that provide insights into startup performance, mentor effectiveness, and overall program outcomes. 7. Ensure a secure system using Supabase Auth with role-based access control and reliable data management via PostgreSQL.

Tech Stack (Software + Hardware)

Frontend — Hosted on Vercel • Framework: Next.js (React-based, supports SSR and static pages) • Language: TypeScript • Styling: Tailwind CSS • UI Components: shadcn/ui • State & Data Fetching: Zustand + React Query • Forms & Validation: React Hook Form + Zod • Charts: Recharts • Deployment: Vercel with automatic CI/CD on every git push Backend — Supabase (fully managed, no custom server needed) • Database: Managed PostgreSQL with auto-generated REST and Realtime APIs • Authentication: Email/password login, OAuth, session management, and role assignment • Row-Level Security: Database-level access policies per user role • File Storage: Built-in storage for pitch decks and documents • Edge Functions: Serverless functions for custom backend logic (e.g. notifications) Database Schema (PostgreSQL via Supabase) Core tables: Users, Startups, Mentors, Programs, Applications, Funding, Events, Sessions, Reports — linked via foreign keys and relational schema.

Methodology / Approach

The platform follows a straightforward two-tier architecture: a Next.js frontend hosted on Vercel communicates directly with Supabase for all backend needs. There is no custom server or middleware layer. When a user opens the platform, the Next.js app loads and authenticates the session using Supabase Auth. Based on the user's role (stored in Supabase as user metadata), the frontend renders only the relevant pages and UI components. All data is read from and written to the Supabase PostgreSQL database using auto-generated Supabase APIs. Row-Level Security (RLS) policies enforced at the database level ensure that each role can only access its own data — for example, a Mentor can only view startups assigned to them, and a Founder can only see their own application and milestones. File uploads (pitch decks, documents) go directly to Supabase Storage with role-based access policies. Real-time subscriptions are used for dashboard widgets, so changes made by one user (e.g. an admin updating a milestone) are reflected instantly without a page refresh. Any backend logic that cannot run on the frontend — such as email notifications or computed aggregates — is handled by Supabase Edge Functions. Layer Technology Responsibility Client (Browser) Next.js + TypeScript Renders pages, manages routing, handles form inputs, displays dashboards and charts UI Layer Tailwind CSS + shadcn/ui Consistent, responsive interface across all stakeholder views State & Data Fetching Zustand + React Query Manages client-side state; caches and syncs server data without manual refetching Authentication Supabase Auth Handles login, registration, password reset, session tokens, and OAuth. Roles (Admin, Mentor, Founder, Investor, Manager) are stored as user metadata Database Supabase PostgreSQL Stores all application data — users, startups, mentors, programs, events, funding, sessions. Tables are linked via foreign keys Row-Level Security Supabase RLS Policies Database-level access rules ensure each role only reads/writes its own data, without any custom middleware Real-Time Updates Supabase Realtime Dashboard widgets subscribe to live database changes — mentor sessions, milestone updates, and funding changes reflect instantly File Storage Supabase Storage Stores pitch decks, program documents, and event materials. Access is controlled per role using storage policies Custom Logic Supabase

Edge Functions Handles any server-side logic that cannot run on the frontend, such as sending email notifications or computing aggregate stats Deployment Vercel Hosts the Next.js frontend with automatic CI/CD on every git push. No server to manage Data flow for a typical action (e.g. Founder submits a startup application): 1. Founder fills the application form on the Next.js frontend. 2. React Hook Form validates the input using Zod schema before submission. 3. The form data is sent to Supabase via the auto-generated REST API. 4. Supabase RLS checks that the user is a Founder and allows the insert. 5. The record is saved in the Startups and Applications tables in PostgreSQL. 6. The Admin's dashboard receives a real-time update via Supabase Realtime subscription. 7. If configured, a Supabase Edge Function sends an email notification to the admin.

Modules / Work Breakdown

1. User Authentication & Role Management • Registration, login, password reset via Supabase Auth • Roles: Admin, Startup Founder, Mentor, Investor, Incubator Manager • Role-based page access enforced on the frontend; data access enforced via RLS 2. Startup Management • Startup application submission and onboarding • Startup profile management (team, idea, sector) • Milestone creation and progress tracking 3. Mentor Management • Mentor profile and onboarding • Mentor-startup assignment by admin • Session scheduling and feedback submission 4. Incubator Program Management • Cohort and program creation by admin • Application review and selection workflow 5. Funding Tracker • Record and track grants, seed funding, and investment rounds per startup • Display funding milestones and current status on startup dashboard 6. Event & Workshop Management • Create and manage training sessions, workshops, and demo days • Event listing and basic registration tracking 7. Analytics Dashboard • Startup growth and milestone completion charts (Recharts) • Program-level overview for admins: cohort status, pending applications, active mentors • Funding progress summary across all startups

Expected Output

The project will deliver a fully functional web-based platform where startup founders can register and track their incubation progress, administrators can manage cohorts, assign mentors, and monitor performance, mentors can schedule sessions and provide feedback, and all stakeholders can view relevant analytics from a single dashboard. The system will improve transparency and reduce manual coordination effort in managing startup incubation programs.

References

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